

TOWARDS CONTROLLABLE PERCUSSION: REVISITING DRUMBLENDER

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ABSTRACT

Differentiable modeling of percussive audio remains highly challenging due to the extreme acoustic variance of transient and decay envelopes across instruments. This paper revisits Drumblender [1], a differentiable analysis–synthesis model with explicit sinusoidal, noise, and transient pathways, and studies two structural modifications for improved reconstruction: a refined CQT-based modal extraction front-end and a DAC-based noise-branch encoder with optional temporal modeling. Experiments show that the DAC-based encoder yields the clearest improvement over the baseline in objective reconstruction quality, while the refined CQT front-end does not improve performance in its current form. The results indicate that branch-specific encoder design is a more promising direction than front-end modal refinement under the present setting, and further underscore the difficulty of robust differentiable modeling for diverse percussive audio.

1. INTRODUCTION

Recent advances in deep learning have significantly expanded the scope of audio modeling and synthesis. Beyond classical physical modeling and digital signal processing [2, 3], data-driven methods have shown strong performance in reconstruction tasks. In particular, differentiable digital signal processing (DDSP) [4] combines interpretable synthesis modules with gradient-based learning, enabling structured analysis–synthesis systems to be trained directly from audio.

Prior work on percussion has developed along several complementary directions, including neural drum generation [5–7], descriptor- or feature-conditioned synthesis [8–11], and latent- or control-oriented analysis of percussive sound models [12, 13]. At the same time, physically grounded studies have examined specific percussive systems, modal behaviors, and transfer characteristics [14–16]. Nevertheless, percussive audio remains challenging because it typically consists of a short, wideband transient followed by a decay whose temporal and spectral characteristics vary widely across instruments, making

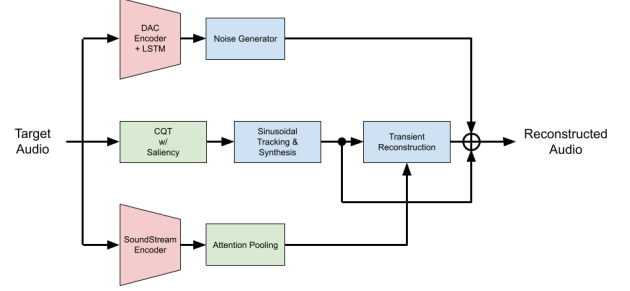


Figure 1. Modified Drumblender configuration used in this work, consisting of the saliency-based CQT sinusoidal branch, the DAC+LSTM noise branch, and the original transient reconstruction pathway, with the $T(S) = S + N$ structure.

it difficult to impose a single physically motivated prior across diverse percussion sources, especially in label-free settings.

Drumblender addressed this problem with a differentiable analysis–synthesis framework composed of sinusoidal, noise, and transient pathways, where a TCN-based transient module improved onset reconstruction for percussive sounds [1]. This formulation provides an interpretable decomposition for percussive audio, but robust reconstruction across diverse and variable signals remains difficult. In particular, modal extraction from CQT peaks can be unstable for dense broadband percussion, and encoder designs inherited from general neural audio codecs are not always well matched to the structural requirements of each synthesis branch. We introduce two primary structural modifications: a refined front-end modal extraction pipeline for robust separation of tonal and chaotic noise components, and revised downstream waveform encoders that better capture the temporal dynamics of each synthesis branch.

2. METHOD

2.1 Refined CQT-Based Modal Analysis

The baseline framework proposed by Shier et al. [1] extracts sinusoidal trajectories via reverse-time peak detection on the Constant-Q Transform (CQT) magnitude spectrogram. While highly effective for sub-bass pitch glides in kicks and toms, this approach struggles with dense broad-

band percussion like cymbals, inadvertently tracking erratic noise spikes as stable sinusoids. Because the architecture’s synthesis branches are interdependent, these hallucinated sinusoids detract from the energy modeled by the noise pathway. We hypothesize that refining the modal branch to reject these artifacts will encourage the noise and transient pathways to better capture their respective targets, ultimately improving overall reconstruction.

To achieve this while preserving CQT’s strengths, we introduce Frequency-Axis Median Saliency prior to peak detection. Given the temporally reversed magnitude spectrogram $\mathbf{X} \in R^{K \times T}$, we estimate a dynamic noise floor $\mathbf{B}$ using a vertical median filter:

$$\mathbf{B}(k, t) = \text{median}_p\{\mathbf{X}(k, t)\} \quad (1)$$

where p is the spectral kernel size. Subtracting this background yields the Saliency map $\mathbf{S}$:

$$\mathbf{S}(k, t) = \max(0, \mathbf{X}(k, t) - \mathbf{B}(k, t)) \quad (2)$$

Since broadband cymbal noise exhibits a relatively flat spectral envelope, it is closely approximated by the median filter. This subtraction efficiently suppresses the noise wash while preserving sharp, isolated tonal peaks. Peak detection is then performed exclusively on the logarithmic Saliency map ($\mathbf{S}_{dB}$). Finally, to ensure accurate downstream additive synthesis, sub-bin frequencies and phases are interpolated using the original magnitude $\mathbf{X}$.

2.2 Encoder Architecture Improvements

The original encoders use a SoundStream-based structure for both the noise and transient branches [17]. For the noise branch, we replace the original encoder with a DAC-style convolutional encoder [18] while preserving the original interface, namely waveform input, total stride of 128, and output dimensionality of 128. Compared with the original SoundStream encoder, the DAC-style encoder adopts Snake1d activations and weight-normalized convolutions, providing a stronger local feature extractor under the same output resolution. To introduce lightweight temporal modeling for the frame-aligned noise-control sequence, a two-layer unidirectional LSTM with hidden size 256 is appended after downsampling, followed by a linear projection back to 128 dimensions:

$$\mathbf{Z}_{\text{noise}} = \text{Linear}(\text{LSTM}(E_{\text{dac}}(\mathbf{x}))). \quad (3)$$

This modification keeps the original noise-synthesis interface unchanged while allowing explicit temporal modeling in the noise encoder.

An overall diagram of the implemented architecture is depicted in Figure 1.

3. EXPERIMENTS

For training, dataset was constructed from commercial or freely available sample packs. Only unclipped one-shot, single-strike samples with sample rates 48kHz were used.

Table 1. Metric table using MSS, LSD, and SF(Spectral Flux Onset Error). Five configurations were evaluated: the variable-length baseline, DAC, DAC+LSTM, Refined CQT, Refined CQT+DAC.

Mode	MSS ↓	LSD ↓	SF ↓
Baseline	1.399 ± 0.301	2.307 ± 0.599	282.687
DAC	1.385 ± 0.278	2.256 ± 0.444	284.596
DAC+LSTM	1.380 ± 0.381	2.258 ± 0.579	407.746
Ref. CQT	1.554 ± 0.800	2.362 ± 0.619	468.392
Ref. CQT+DAC	1.419 ± 0.328	2.307 ± 0.514	327.927

The dataset included both acoustic and electronic percussive sounds with variation in strike intensity. Samples with structurally complex or heavily processed content, such as mixtures of multiple sound sources, moving stereo panning, or strong delay and reverberation effects, were excluded. In total, approximately 13,000 samples were used, with an 80/10/10 train/validation/test split balanced at the sample-pack level.

All models were trained using AdamW with an initial learning rate of 1×10^{-4} and weight decay of 5×10^{-7} . All experiments used the transient-enabled Drumbler setup $T(S) = S + N$. The baseline model used the original MR-STFT loss, with variable-length loss computed only over the valid unpadded region.

4. RESULTS

Table 1 summarizes the objective evaluation results. The baseline remained a stable reference despite the lighter overall training setup. Among the test results, replacing the noise encoder with a DAC-based encoder yielded the clearest improvement over the baseline, while the additional LSTM produced only mixed gains. The refined CQT front-end did not improve reconstruction quality in its current form, and although combining it with the DAC encoder partially recovered the degradation, it did not surpass the simpler DAC-based variant.

5. DISCUSSION AND FUTURE WORK

This work presented structurally informed modifications to the differentiable Drumbler framework, introducing a frequency-axis median saliency pipeline for modal analysis, DAC-based encoder revisions, and variable-length training support. While noise branch encoder upgrades yielded objective improvements, the integration of LSTM-based temporal modeling and variable-length configurations produced mixed or degraded results. Although the refined CQT-based modal extraction pipeline demonstrated perceptually cleaner separation when evaluated in isolation, its deployment within the joint training framework unexpectedly degraded overall objective metrics.

Our immediate future work will focus on addressing these temporal modeling bottlenecks, investigating more stable conditioning mechanisms for variable-length generation without sacrificing reconstruction fidelity. Beyond improving reconstruction, the ultimate goal of this project

aligns with the original vision of differentiable analysis–synthesis: to build a highly controllable and interpretable latent representation for percussion. By ensuring a clean decomposition of sinusoidal, noise, and transient components, we aim to unlock advanced downstream applications such as percussive audio inpainting, semantic timbre manipulation, and fine-grained transient shaping.

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